

1-3-1: Establishing and Testing Theories in Troubleshooting

At the end of this episode, I will be able to:

1. Identify and explain the significance of establishing and testing probable theories, given a scenario.

Learner Objective: *Identify and explain the significance of establishing and testing probable theories, given a scenario*

Description: In this episode, the learner will explore establishing and testing theories when applying a troubleshooting methodology. We will explore some techniques, such as questioning the obvious or simple fixes first and considering multiple theories, such as using the OSI model to approach the theory creation process.

- Can you describe the what we mean when we say "Establishing and Testing Theories in Troubleshooting?"
 - o Formulating hypotheses about the root cause
 - o systematically test these theories to narrow down the possibilities.
- What are some of the techniques that we can utilize to come up with possible theories?
 - o Question the Obvious
 - ♣ Start by questioning basic assumptions
 - ♣ Verifying simple configuration
 - ♣ Try to ensure that no details have been overlooked
 - o Consider Multiple Approaches:
 - ♣ Top-to-bottom/bottom-to-top (OSI Model)
 - ♣ Approach the problem by working through the OSI model layers
 - ♣ From the Physical Layer or Layer 1 up to the Application layer or Layer 7 (bottom-to-top)
 - ♣ From the application layer down (top-to-bottom)
 - ♣ Done to isolate the issue or issues systematically.
 - ♣ Divide and Conquer
 - ♣ Break down the problem into smaller, manageable segments
 - ♣ Helps to identify the root cause more efficiently
 - ♣ Focuses on isolating specific areas or functions where the issue could reside.
- Now that we have some theories, what do we do with them next?
 - o Test the theory to determine the cause - practical experimentation and observation to validate or refute the established hypothesis or theory.
- What if we confirm our theory to be correct?:
 - o If the theory has been confirmed
 - ♣ Determine the next steps to fully resolve the problem. This may include:
 - ♣ Applying fixes
 - ♣ Changing settings,
 - ♣ Replacing faulty hardware.
- What if we find our theory to be incorrect?
 - o If the theory is not confirmed
 - ♣ Reassess the situation to establish a new theory based on the information gathered from the test
 - ♣ Consider alternative explanations for the issue, leading to a fresh set of theories
- What if we find that the issue is alluding us, or beyond the scope of our expertise, what do we do?
 - o We escalate the problem - Escalation - escalate the issue to a higher-level support team or specialist with more expertise or resources.
 - o We can escalate the show escalated the
 - ♣

- If after several attempts the problem remains unsolved
 - ♠ If the complexity of the issue is beyond the current skill level
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- Additional Resources:
 - o If applicable